

ALPACA OPTIONS ALPHA AGENTS HACKATHON

Options Sniper

Autonomous Bull Call Spread Paper-Trading Agent

Dynamic Market Discovery · Options Intelligence · News Risk Assessment

Serverless Next.js 15 + Tailwind · Alpaca Trading & Market Data APIs · Zero Database

Paper Trading Only — No Real Money · Built for the Alpaca Hackathon

Architecture Overview

TECH STACK

Framework: Next.js 15 (App Router) **Styling:** Tailwind CSS 4

Hosting: Vercel (Serverless) **Data:** Alpaca APIs Only

Database: None (stateless) **Auth:** Demo (paper-only)

DESIGN PRINCIPLES

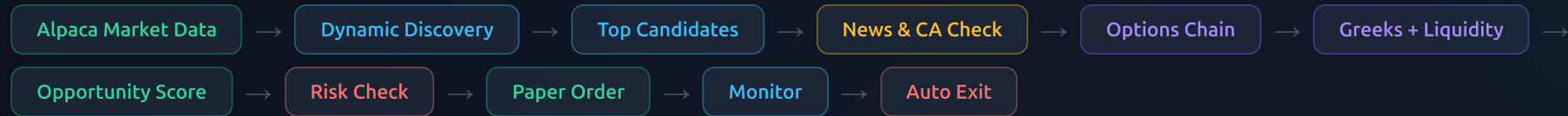
- **Alpaca is the single source of truth** — no database, no caching
- **Serverless API routes** — each tick is a stateless function call
- **Client-driven polling** — the browser drives the 60s agent loop
- **Paper trading hard guard** — live URL is blocked at code level
- **Full mock mode** — demo without Alpaca keys via `ALPACA MOCK=true`

PROJECT STRUCTURE

```
app/  
  page.tsx           ← Dashboard UI  
  login/page.tsx     ← Demo auth  
  components/  
    AlpacaMonitor.tsx ← Live API activity  
    MarketDiscovery.tsx ← Top movers panel  
    OptionsIntelligence.tsx ← Greeks/IV panel  
    NewsRisk.tsx      ← News & CA risk  
    ui.tsx            ← Shared components  
  api/  
    agent/tick/route.ts ← Full autonomous tick  
    agent/scan/route.ts ← Scan only (no trade)  
    account/route.ts   ← Account + positions  
    trade/route.ts     ← Manual trade submit  
    positions/close/   ← Close spread  
    alpaca/monitor/    ← API activity log  
  
lib/  
  alpaca.ts          ← REST client (server-only)  
  agent.ts           ← Agent orchestrator  
  scanner.ts         ← Scan → indicators → spreads  
  discovery.ts       ← Dynamic market discovery  
  intelligence.ts    ← Options Greeks/IV/liquidity  
  newsRisk.ts        ← News sentiment & CA risk  
  indicators.ts      ← SMA, RSI, momentum  
  mock.ts            ← Deterministic demo data  
  apiMonitor.ts      ← In-memory API logger  
  types.ts           ← Shared TypeScript types
```

⚙️ Autonomous Agent Pipeline

Each agent tick executes the complete flow below — from market data to paper order — in ~10-30 seconds.



📊 DATA INPUT

Screener: Top gainers, losers, most active

Bars: 70-day daily OHLCV

Options: Call snapshots 14-45 DTE

News: Headlines + keyword sentiment

Corp Actions: Dividends, splits, mergers

🧠 ANALYSIS

Indicators: SMA(20), SMA(50), RSI(14), 5-bar momentum

Trend: Bullish = price > MA20 > MA50

Greeks: IV, Delta, Gamma, Theta, Vega

Quality: Options quality 0-100 composite

Risk: News impact ± 5 , CA warnings reject

📈 OUTPUT

Score ≥ 75 + Risk $\leq 1\%$: Auto-enter spread

Position sizing: Max 1% equity per trade, cap 10

Exit rules: +25% profit, -20% loss, ≤ 7 DTE, bearish reversal

Order type: Multi-leg (mleg) limit with slippage

Close order: Short leg first, then long (avoid naked short 403)

Feature 1: Dynamic Market Discovery

Replaces fixed-symbol scanning with real-time Alpaca screener data to find the best candidates every tick.

ALPACA API ENDPOINTS

GET `/v1beta1/screener/stocks/movers?top=10`

Returns top gainers + losers with change % and price

GET `/v1beta1/screener/stocks/most-actives?by=volume&top=10`

Returns most actively traded stocks by volume

GET `/v1beta1/options/snapshots/{symbol}?feed=indicative&type=call`

Used to verify options availability for each mover

QUALIFICATION PIPELINE

- Fetch movers (gainers, losers, most active)
|
- Check options availability per symbol
|
- Filter: losers rejected (Bull Call Spread only)
|
- Filter: low momentum (<1%) rejected
|
- Merge with default watchlist, cap at 15 candidates

DISCOVERY OUTPUT EXAMPLE

```
NVDA +5.2% → Options Available → ✓ Qualified
AMD +4.7% → Options Available → ✓ Qualified
META +3.8% → Options Available → ✓ Qualified
AVGO +3.1% → Options Available → ✓ Qualified
PLTR +2.9% → Options Available → ✓ Qualified
AAPL +1.1% → Options Available → ✓ Qualified
COIN -4.2% → Bearish → ✗ Rejected
RIVN -3.8% → Bearish → ✗ Rejected
INTC -2.1% → Bearish → ✗ Rejected
TSLA 120M vol → Active → ✓ Qualified
```

SMART CANDIDATE SELECTION

- **Dynamic first:** Top qualified gainers/active sorted by change %
- **Watchlist fallback:** Default symbols (SPY, QQQ, NVDA...) always included
- **Concurrency control:** Scans up to 5 symbols in parallel
- **Deduplication:** Same symbol across categories merged, gainer priority



Feature 2: Options Intelligence

Deep analysis of each candidate's options chain using Alpaca's Greeks, IV, and market data — no client-side Black-Scholes needed.

ALPACA OPTIONS SNAPSHOT DATA

GET `/v1beta1/options/snapshots/{underlying}?feed=indicative`

Returns greeks, impliedVolatility, dailyBar (volume), latestQuote, OI per contract

GREEKS PARSED FROM ALPACA

Delta: Option price sensitivity to \$1 underlying move

Gamma: Rate of change of delta

Theta: Daily time decay (negative for longs)

Vega: Sensitivity to 1% IV change

IV: impliedVolatility (top-level field, annualized)

Volume: dailyBar.v — contracts traded today

OPTIONS QUALITY SCORE (0-100)

Bid/Ask Spread		40 pts
Open Interest		30 pts
Option Volume		20 pts
IV Sweet Spot		10 pts

INTELLIGENCE PANEL EXAMPLE

QQQ 480/495C 2026-10-16 (48 DTE)

Greeks:

IV: 34.2%
Delta: 0.58
Gamma: 0.0210
Theta: -0.0823
Vega: 0.481

Market Data:

Option Volume: 4,821
Open Interest: 18,432
Bid/Ask Spread: 3.1%
Strike Distance: 1.2% OTM

Ratings:

Liquidity: GOOD
Volatility: ELEVATED
Quality: 87/100

RATING DEFINITIONS

GOOD liquidity: Bid/ask spread $\leq$ 5%

FAIR liquidity: Spread 5-15%

Feature 3: News & Corporate Action Risk Check

Uses Alpaca's news and corporate-action APIs with deterministic keyword sentiment — no external AI/LLM needed.

ALPACA API ENDPOINTS

GET `/v1beta1/news?symbols={sym}&limit=10&sort=desc`

Returns headlines, source, timestamps — no built-in sentiment

GET `/v1/corporate-actions?symbols={sym}&start=...&end=...`

Grouped response: cash_dividends, forward_splits, cash_mergers, etc.

DETERMINISTIC SENTIMENT CLASSIFICATION

Positive keywords: beat, exceed, outperform, upgrade, bullish, rally, surge, record, strong, FDA approval...

Negative keywords: miss, disappoint, downgrade, bearish, crash, lawsuit, SEC probe, recall, bankruptcy, warning...

Earnings detection: earnings, EPS, quarterly results, guidance, 10-Q/K, conference call...

RISK ASSESSMENT EXAMPLE

NVDA

Recent News:

- ▲ NVIDIA beats Q2 earnings expectations (Reuters)
- ▲ NVIDIA announces next-gen GPU architecture (Bloomberg)
- ▲ Analysts raise NVIDIA price targets (MarketWatch)

News Sentiment: Positive

Corporate Actions: Clear

News Impact: +4

Risk Warning: None

AAPL

Recent News:

- ◆ Apple announces new iPhone lineup (TechCrunch)
- ▲ Apple Services revenue hits all-time high (WSJ)

Corporate Actions:

- ▲ Cash Dividend \$0.25/share ex-date in 12 days

News Sentiment: Positive

Corporate Actions: Warning

News Impact: +1

Risk Warning: None (dividend is routine)



Alpaca Paper Trading Integration

Every trade is a multi-leg options order on Alpaca's paper trading API. Positions are monitored and automatically exited.

TRADING API ENDPOINTS

GET /v2/account

GET /v2/positions

GET /v2/orders?status=open&limit=50

POST /v2/orders (mleg multi-leg spread)

DEL /v2/orders/{id} (cancel stale orders)

DEL /v2/positions/{symbol} (close legs)

ORDER CONSTRUCTION

```
{
  "order_class": "mleg",
  "qty": "3",
  "side": "buy",
  "type": "limit",
  "limit_price": "4.55",
  "time_in_force": "day",
  "legs": [
    { "symbol": "NVDA261016C00128000",
      "qty": "1", "side": "buy",
      "type": "limit", "limit_price": "4.55",
      "time_in_force": "day" },
    { "symbol": "NVDA261016P00128000",
      "qty": "1", "side": "sell",
      "type": "limit", "limit_price": "4.55",
      "time_in_force": "day" }
  ]
}
```

Options Sniper by Alpaca Options Alpha Agents

BULL CALL SPREAD CONSTRUCTION

Long leg (ATM): Strike closest to current price

Short leg (OTM): $\geq 3\%$ above long; delta ≈ 0.30 target

DTE window: 14-45 days to expiration

Liquidity filter: Bid > 0, Ask > 0, Spread $\leq 25\%$, OI ≥ 100

Reject: Debit $\leq \$0.15$, debit $\geq$ width, risk/reward < 0.8

Slippage: \$0.05 per-share cushion on limit price

AUTO-EXIT RULES

+25% Profit

Take profit at 25% gain on debit

-20% Max Loss

Cut loss at 20% of debit paid

≤ 7 DTE

Time exit — close before theta decay accelerates

Bearish Reversal

Underlying price < MA20 < MA50

CLOSE ORDER SEQUENCE

Short leg first → Long leg second

Prevents naked short option 403 errors on level-3 accounts

Opportunity Scoring System (0-120)

Every candidate spread receives a composite score from 7 factors. Score ≥ 75 with risk $\leq 1\%$ equity triggers an automatic entry.

BASE SCORE (0-100)

Trend  25

MA separation rewards: wider 20MA vs 50MA gap = higher score

RSI-14  20

Sweet spot 50-65 (bullish not overbought). $>72 = 0$ (overbought)

Momentum  20

5-bar rate of change. $+2.5\%$ or more = full 20 points

Liquidity  15

Open interest depth + bid/ask spread tightness

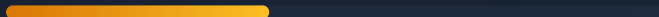
Risk/Reward  20

Risk:Reward $\geq 2:1$ = full 20 points

ENHANCED SCORE (0-120)

Options Quality  +10

Spread 40pts + OI 30pts + Volume 20pts + IV sweet spot 10pts $\rightarrow$ scaled to 0-10 bonus

News Impact  ± 5

-5 (severe negative + CA warning) to +5 (strong positive news)

Impact $\leq -5 \rightarrow$ trade rejected entirely

SCORE CALCULATION EXAMPLE

NVDA 128/133C 2026-10-16 (48 DTE)

Trend: 22.5 (strong bullish, MA gap 1.2%)

RSI: 18.0 (RSI 58, sweet spot)

Momentum: 17.5 ($+2.1\%$ 5-bar ROC)

Liquidity: 12.0 (OI 12K, spread 3.2%)

Risk/Reward: 16.0 (RR 1.6:1)

Options Q: 8.7 (quality 87/100)

News: +2.0 (positive earnings, no CA risk)

TOTAL: 96.7 / 120 $\checkmark$ ENTER (≥ 75 threshold)

Alpaca Integration Monitor

Real-time view of every Alpaca API call — endpoints, latency, status, and sanitized response bodies. No secrets ever reach the browser.

API ACTIVITY TRACKING

Every call through the **request()** wrapper in `lib/alpaca.ts` is logged to an in-memory ring buffer (250 entries).

- **Latency:** Measured with `performance.now()` per call
- **Category:** account | market | options | trading
- **Status:** HTTP status code + ok/fail
- **Response:** Sanitized 700-char JSON snippet (no keys/secrets)
- **Mock flag:** Demo calls tagged separately

CATEGORIES & NEW ENDPOINTS

MARKET	Bars, quotes, screener movers, news, corporate actions	OPTIONS	Chain snapshots, Greeks, IV, dailyBar volume
TRADING	Orders (mleg), positions, cancel, close	ACCOUNT	Equity, buying power, status

7-STEP AGENT PIPELINE (MONITOR VIEW)

1. **AGENT reads account** — GET `/v2/account`
|
2. **ALPACA market discovery** — screener/movers + most-actives
|
3. **ALPACA news & corp actions** — news + corporate-actions
|
4. **ALPACA options chain** — snapshots + Greeks + IV
|
5. **AGENT decision** — Score → ENTER / WAIT / EXIT
|
6. **ALPACA paper order** — POST `/v2/orders` (mleg)
|
7. **ALPACA monitor positions** — GET `/v2/positions`

TRANSPARENCY

Every scan clearly shows **which Alpaca data was requested** and **how it influenced the final trading decision** — from movers qualification through Greeks analysis to news risk impact.

Complete Alpaca API Map

Every Alpaca endpoint used by Options Sniper, organized by function. All calls go through the monitored request() wrapper.

ACCOUNT

GET /v2/account

Equity, buying power, status

MARKET DATA

GET /v1beta1/screener/stocks/movers

GET /v1beta1/screener/stocks/most-actives

GET /v2/stocks/{sym}/bars

GET /v2/stocks/quotes/latest

GET /v1beta1/news

GET /v1/corporate-actions

OPTIONS DATA

GET /v1beta1/options/snapshots/{underlying}

Greeks (δ , γ , θ , ν), IV, dailyBar volume, OI, bid/ask

Feed: indicative (free tier, 15min delayed)

Paginated: up to 4×500 results

Filtered: type=call, 14-45 DTE window

TRADING

GET /v2/positions

GET /v2/orders?status=open

POST /v2/orders (mleg)

DEL /v2/orders/{id}

DEL /v2/positions/{symbol}

13

API ENDPOINTS

4

CATEGORIES

2

BASE URLS

0

EXTERNAL AI APIS

Options Sniper

Autonomous Bull Call Spread paper-trading agent powered entirely by Alpaca APIs



DYNAMIC DISCOVERY

Real-time screener for top gainers, losers & most active stocks with options qualification



OPTIONS INTELLIGENCE

Greeks, IV, liquidity ratings & 0-100 quality score from Alpaca's options snapshots



NEWS & RISK CHECK

Deterministic sentiment + corporate action warnings — no external AI needed

PAPER TRADING ONLY

ZERO DATABASE

13 ALPACA ENDPOINTS

NO EXTERNAL AI

FULL API MONITOR

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